AI FAQ & Customer Support Automation

API TESTING REPORT SPECIFICATION

TESTING OBJECTIVE

Verify authentication, protected endpoints, FAQ and ticket management/search, and Gemini AI endpoints, confirming structured JSON responses and error handling.

Environment: Node.js + Express.js; MongoDB; Thunder Client in VS Code / Postman; Bearer JWT; Google Gemini API; Local server `http://localhost:5000`.

Authentication Tests

- `POST` `/api/auth/register` — valid user/agent creation.
- `POST` `/api/auth/login` — valid credentials & JWT.
- `GET` `/api/auth/profile` — valid Bearer JWT.
- **Negative tests:** invalid credentials, missing/invalid authorization tokens.

Support & FAQ Management Tests

- `POST` `/api/tickets` — create valid ticket.
- `GET` `/api/tickets` — list user/agent tickets.
- `GET` `/api/tickets/:id` — retrieve one.
- `PUT` `/api/tickets/:id` — update status/priority.
- `DELETE` `/api/tickets/:id` — delete ticket.
- `GET` `/api/faqs/search` — search by title/category.
- **Validation:** required fields, enum status/priority bounds.

AI INTEGRATION TESTS

`POST` `/api/ai/faq-query`

Input: `userQuery`, `category`, `contextHistory`

Expected Output: Context-aware answer, matched FAQ links, confidence score, and suggested follow-ups.

`POST` `/api/ai/ticket-auto-reply`

Input: `ticketDescription`, `userAccountTier`, `category`

Expected Output: Auto-reply draft, intent tag, sentiment analysis, priority ranking, and escalation alert.

EXPECTED RESULTS & VERIFICATION

Successful requests return `2xx` JSON responses. Authentication, validation, database, and internal failures return structured error JSON. Evidence includes screenshots for registration, login, profile, ticket CRUD/search, automated query resolution, and AI support insights.